

Program : Diploma in Robotics and Automation	
Course Code : 2379	Course Title: Foundations of Robotics Lab
Semester : 2	Credits: No Credit
Course Category: Engineering Science	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To create the skill that is required to develop simple hobby projects using electronic components, Arduino, Raspberry pi, and compatibles which does not envisage prior knowledge of C or Python, instead, students are exposed to the real-time learning experience which helps students to better understand programming languages.

General Instructions:

Tinkering is about creating interest among students in making. Electronics Tinkering Workshop begins with soldering practice. Since the safety of the student is of prime consideration, usage of fume extractors and proper soldering iron stand, etc are recommended. A necessary antistatic workbench for ESD safety shall be provided. Faculty shall ensure proper usage of soldering tools to ensure the quality of soldering and safety of components.

Course outcomes CO3 and CO4 are set to familiarize Arduino and Raspberry pi through a real-time learning experience. The course is an attempt to kindle interest among students in coding through the live experience. The majority of the course contents encourage self-learning and learning through experience. The course does not envisage prior knowledge in any programming language. The faculty need not attempt to teach students C or python programming and most of the codes are available in the example library of the IDE or can be downloaded from GitHub. It is only required to make necessary modifications to suit the code for matching with the problem statement. The students may be instructed to follow coding standards as followed in the example library. This shall include the insertion of proper comments, acknowledgment of copyright, etc.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic concepts of computer programming.		Introduction to IT systems Lab	1

Tools and Equipment:

1. Antistatic soldering workbench.
2. Arduino, Arduino shields and compatibles.
3. Computer systems for Arduino IDE.

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	To explore the application and practical dimensions of robots.	9	Applying
CO2	To familiarize with various robotic sensors and their applications	9	Applying
CO3	Develop simple interfacing applications using Arduino.	15	Applying
CO4	Develop simple autonomous robots using Arduino	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Robotic Case Study and Disassembling		
M1.01	Identify a biomedical application of robot and present a case study	1	Applying

M1.02	Identify a space application of robot and present a case study	1	Applying
M1.03	Identify a defense application of robot and present a case study	1	Applying
M1.04	Identify a domestic (like Alexa) application of robot and present a case study	1	Applying
M1.05	Identify an industrial application of robot and present a case study	1	Applying
M1.06	Identify a robotic application in automobiles and present a case study	1	Applying
M1.07	Disassemble an autonomous robot and identify the parts	3	Applying
CO2	To familiarize with various robotic sensors and their applications		
M2.01	Study of the characteristics of a Photo reflective sensor for Distance Measurement	1.5	Applying
M2.02	Study of the characteristics of a Photo reflective sensor for Speed Measurement	1.5	Applying
M2.03	Study of the characteristics of a Magnetic Proximity sensor for Speed Measurement	1.5	Applying
M2.04	Study of the characteristics of a Thermocouple	1.5	Applying
M2.05	Study of the characteristics of an LVDT	1.5	Applying
M2.06	Study of the characteristics of a bio-sensor	1.5	Applying
	Lab Exam I	1.5	
CO3	Develop basic interfacing applications using Arduino.		
M3.01	Familiarize Arduino UNO, Arduino IDE	1.5	Understanding
M3.02	Blink LED connected to P13 at 1 sec interval. Interface relay module with port and switch relay on and off at different intervals.	1.5	Applying
M3.03	Connect PIR sensor (motion sensor) to port and switch relay connected to port on detecting motion.	1.5	Applying

M3.04	Familiarize DC motor control shield and control DC motor for forward movement, reverse movement and stop.	1.5	Applying
M3.05	Use wheeled robot kit (containing two geared DC motors, caster wheel, Arduino uno, motor driver shield, 9V battery and line sensor) to make line follower robot.	3	Applying
M3.06	Familiarize blue tooth module (HC 05 or HC 06) to switch relay connected to port using smart phone.	1.5	Applying
M3.07	Use wheeled robot kit and blue tooth module to control robot using smart phone.	1.5	Applying
M3.08	Use LCD shield and display your name and class number.	1.5	Applying
M3.09	Implement suitable Do it Yourself (DIY) project from internet and implement using Arduino and compatibles. (Open ended experiment)	1.5	Applying
CO4	Develop autonomous robots using Arduino		
M4.01	Controlling DC Motors with the L298N Dual H-Bridge and an Arduino	2	Applying
M4.02	Line Follower Robot using Arduino	3	Applying
M4.03	Obstacle avoiding robot using Arduino	4	Applying
	Lab Exam II	1.5	

Online Resources:

Sl.No	Website / Link
1	https://www.arduino.cc/
2	http://sl-coep.vlabs.ac.in/
3	https://www.instructables.com/
4	https://github.com/